



## EXAMINERS' REPORT ON 2007 TERTIARY ENTRANCE EXAMINATION

### SUBJECT: PHYSICS

#### *Statistics*

| Year | Number who sat | Non-examination candidates | Absent |
|------|----------------|----------------------------|--------|
| 2007 | 3020           | 27                         | 142    |
| 2006 | 2778           | 22                         | 159    |
| 2005 | 3021           | 29                         | 170    |

This examiner's report is written by the chief marker in collaboration with the chief examiner to comment on matters relating to the Physics Tertiary Entrance Examination. The opinions and recommendations expressed in this report are those of the chief marker and examiner and not necessarily representative of, or endorsed by, the Curriculum Council.

The marking guideline provided at the end of this report was prepared for markers and has been substantially amplified by discussions held in the pre-marking meeting. It is not intended as a set of model answers, and is not exhaustive as regards alternative answers. Some of the answers are less than perfect, but represent a standard response that the examiners deemed sufficient to earn full marks. Teachers who use this guide should do so with its original purpose in mind.

#### *Summary*

The markers felt that the paper gave good, clear instructions to the candidates with the early part of most questions in section B accessible to all candidates and the later parts becoming more difficult. A variety of question styles was used. The marking of the paper was mostly straightforward with markers finding it easier to mark than the 2006 paper. The markers also felt that the paper was less discriminatory at the top end which seems justified with over 36 candidates gaining marks between 93% and 98%.

The format of the paper was the same as last year. The majority of the candidates appeared to complete the paper in the time allowed. The 2007 mean raw exam score of 62% was considerably higher than that in 2006 (55%), and the standard deviation of 18.3 was also higher than that of 2006 (16.5). This suggests an easier paper than last year with a greater spread of marks. This was reflected in the maximum mark of 98% in 2007 compared with 92% in 2006. The range went from 0 to 98%.

Significant figures were again an issue with many candidates using an inappropriate number and this continues to be a worsening problem. Candidates are penalised on this in all questions that require an estimate but this is only to obtain consistency in marking.

#### *General comments*

Firstly thanks must go to the examining panel for their dedication in producing a paper that was well received by candidates, teachers and markers. The major criticisms were:

1. that there was not enough space left below some questions to adequately cater for the different size writing styles or for a more complete answer.
2. that there was again no extra paper in the booklet if candidates needed more

space or had to redo a question.

3. that the graph paper was not printed in the examination booklet. This also occurred two years ago.

These first two points were criticisms of last year's layout.

The design of the paper this year was consistent with the format used in 2006. The 2006 paper was considered a little too hard and it was intended that the level of difficulty be reduced for this paper. This was achieved but perhaps to too great an extent with many good candidates and a number of markers feeling that the paper was too easy without enough challenging questions. Few general conceptual problems were encountered; the glaring exception being back emf in motors. Very few candidates were aware of this and even those that were could not explain it well. The use of diagrams in explanations continues to be poor (These are often essential in order to clarify understanding or provide correct working.), and the ability to derive a given equation using clear steps is generally lacking.

1. There was a deliberate attempt to vary the question types and thus the way in which information was presented to candidates and also in the way in which candidates were required to answer the question. A brief description of the types of question formats used in the paper is given in the section titled "Breakdown of Paper".
2. Section A only contained one item (Q6) this year in which a multiple choice part was used, but for which the candidate was also asked to explain their choice of answer.
3. As in the past two years, there was no choice of alternate-context questions but there was an opportunity in Q9A which offered candidates the choice of answering in relation to specific contexts.
4. Section B again contained a set of questions that provided the candidates with a variety of methods in which to respond (descriptive, graphically, mathematically) and this should be retained for future years. Perhaps questions 1, 3 and 4 were a bit repetitive in parts with effectively the same calculations used.
5. In parts B and C, most questions had an initial part to assist all candidates in starting the question and to focus their attention on the stimulus material provided. This was done in order to reduce the likelihood of them misinterpreting the question.
6. There was a deliberate attempt to include questions or part-questions that specifically tested candidates' conceptual physics understanding.
7. As in 2006, a question on experimental design and the analysis of data was included in Section C rather than Section B or Section A. Due to the varying responses in previous years between those candidates who preferred to formulate a solution using manual graphing to those who preferred to use a graphics calculator, this choice was not available this year with all candidates directed to use a graph. Some candidates still chose to determine the gradient using their graphics calculator.
8. The second question in Section C involved the comprehension of the more complex physics concept of tides and the interpretation of graphical data. Candidates had to read graphs, do simple calculations and extract information. These questions provided a good mix of skills.

### Breakdown of Paper

This breakdown is not always easy to ascertain because many questions, particularly in Parts B and C, cover several areas and/or contexts.

1. The approximate mark allocation for different answer types is given below. The figures are percentages.

|                               | Part A | Part B | Part C | Total |
|-------------------------------|--------|--------|--------|-------|
| Numerical/calculation answers | 15.5   | 36     | 12     | 63.5  |
| Written answers/explanations  | 11     | 12.5   | 4      | 27.5  |
| Graphical or diagrammatic     | 3      | 1.5    | 2      | 6.5   |
| Multiple choice answers       | 0.5    | 0      | 2      | 2.5   |
|                               | 30     | 50     | 20     | 100   |

2. On setting the paper, the breakdown by area of study in comparison with the syllabus requirements is given below. Figures for other papers back to the 2004 paper are also provided. The figures are percentages.

| Topic                    | Syllabus requirement | 2007 paper | 2006 paper | 2005 paper | 2004 paper |
|--------------------------|----------------------|------------|------------|------------|------------|
| Sound Waves              | 15-20                | 15.5       | 18         | 18.5       | 15         |
| Electric power           | 20-30                | 20.5       | 21         | 21         | 30         |
| Movement                 | 20-30                | 28.5       | 27.5       | 20         | 23         |
| Structures and Materials | 15-25                | 18         | 16.5       | 21         | 20         |
| Atomic Physics           | 10-20                | 17.5       | 17         | 19.5       | 12         |

3. Breakdown by parts of the paper was:

|                          | Part A | Part B | Part C | Total |
|--------------------------|--------|--------|--------|-------|
| Sound Waves              | 2      | 13.5   | 0      | 15.5  |
| Electric power           | 12     | 8.5    | 0      | 20.5  |
| Movement                 | 6      | 12.5   | 10     | 28.5  |
| Structures and Materials | 6      | 2      | 10     | 18    |
| Atomic Physics           | 4      | 13.5   | 0      | 17.5  |

## Comments on specific questions

These comments are offered by the Chief Marker after consultation with selected individual markers and the Chief Examiner. Total number of candidates was 3020.

### SECTION A: Short Answer

- Q1                      Attempted      3006                      Mean      73%
- As the first question on the paper it should be one of the easier questions to do and this proved to be the case. Nearly all candidates understood that the velocity was a tangent to the path, although some lost marks due to poorly drawn vectors. The acceleration, which is constant and due to gravity, was not so well done with a number of candidates showing this as upwards in the first part of the path. A few candidates also drew the acceleration perpendicular to the path confusing this with circular motion.
- Q2                      Attempted      2921                      Mean      42%
- This proved to be one of the harder questions on the paper but this was due to the fact that it required candidates to clearly explain the physics behind a simple everyday situation, which normally proves more challenging. Nearly all candidates understood that the situation involved a difference in torque but answers were often incomplete and some did not even refer to the diagram despite instructions to do so. Others complicated the problem by considering angles rather than the perpendicular distances which often lead to contradictions in their arguments.
- Q3                      Attempted      2966                      Mean      75%
- Another simple torque problem but this time requiring a calculation. The main difficulty that candidates had was in understanding and using the concept of centre of mass. Candidates had many strange ideas about splitting up the mass of the board. The other error that was common was taking moments about the wrong point and ignoring the reaction at the end of the wall
- Q4                      Attempted      2965                      Mean      73%
- Although this was another simple calculation candidates lost marks because they could not convert units properly, with the most common error being not adding the altitude to the radius of the Earth. The most common method involved using  $v = 2\pi r/T$  but a number chose the alternative method of using  $v = \sqrt{GM/r}$ .
- Q5                      Attempted      2954                      Mean      85%
- This question proved to be the easiest on the paper with candidates familiar with using  $E = Blv$ . The main error was not calculating a vertical component properly.
- Q6                      Attempted      3007                      Mean      61%
- Nearly all candidates chose Figure B but explanations were often incomplete. The best answers used a field diagram with some excellent answers discussing the torque created by the magnetic field on the compass needle.
- Q7                      Attempted      2886                      Mean      62%
- Many candidates were confused by this question and did not pick up on the fact that the flux cut was twice the maximum as the field reverses through the coil. The majority used the formula  $\Phi = BA$  successfully with the main errors occurring in converting units correctly when calculating the area.

- Q8                      Attempted      2803              Mean      48%  
This had the second lowest attempt rate on the paper and the majority who attempted the question had difficulty in distinguishing the two different waves. Extracting information from the graph was poorly done but most candidates understood that the frequency was the inverse of the period.
- Q9                      Attempted      2906              Mean      63%  
Diffraction was the most common property chosen to illustrate wave like behaviour of light but many did not explain the phenomenon using a clear context. Fewer candidates chose the obvious phenomenon of the photoelectric effect to illustrate the particle nature of light which was surprising. The main marks lost were due to poor explanations of the chosen phenomenon with many weaker candidates simply stating the phenomena.
- Q10                     Attempted      2975              Mean      61%  
A surprising number of candidates cannot distinguish between a motor and a generator. Many candidates who stated it was a motor then gave descriptions of the features which make it a generator. The most common error was not picking up on the fact that the output was AC.
- Q11                     Attempted      2945              Mean      50%  
This was a good discriminator between the strong and weak candidates. Space to answer the question adequately was an issue here with most candidates having to continue answers somewhere else on the page. Candidates had trouble clearly explaining why the emf reverses often not relating concepts such as Lenz's Law to the question. Part (b) was usually answered better with better candidates relating the increase in speed to the rate of change of flux.
- Q12                     Attempted      2848              Mean      43%  
This proved to be a difficult question despite a strong hint given in the question. Many candidates found the speed at the top of the loop which demonstrated a good understanding of vertical circular motion but then failed to correctly apply the principle of conservation of energy. A large number of candidates found the change in height to be 0.2 m but failed to understand that this must be above the top of the loop. To state an answer which had the height of release well below the top of the loop was very disheartening as it shows that a vast number of candidates are not thinking about the Physics of the problem but rather fixated on plugging numbers into a formula.
- Q13                     Attempted      2840              Mean      66%  
Candidates generally answered this question very well or had no idea with little in between.
- Q14                     Attempted      2807              Mean      67%  
Many candidates obviously had this derived formula on their calculator and the question simply became a case of plugging in numbers. For those that had to derive the relationship of  $\tan \theta = v^2/rg$  the question was quite difficult. The major problem was identifying the correct angle in vector diagrams. Candidates who had major difficulties did not draw adequate diagrams.
- Q15                     Attempted      2689              Mean      13%  
This was supposed to be a difficult question and proved to be even harder than expected. It would appear that few candidates have come across the concept of back emf in motors and many of those that have could not adequately explain what was happening in the motor. It was very disconcerting to see so many candidates talk about the current "building up" in the motor.

## SECTION B: *Problem Solving*

- Q1                      Attempted      2983              Mean      66%
- Although this was intended to be a fairly straight forward question many candidates found the diagram confusing and tried to use vectors to find the forces in the frame rather than the arms. Candidates often forgot to take into consideration that there were two arms or forgot to add the mass of the arms to the mass of the baby in calculations. As is often the case in a real life situation, estimations and approximations have to be used and many candidates had trouble estimating the velocity of the swinging arms using the data provided so that a centripetal acceleration could be found. The last part had many factors to consider in calculating a breaking stress and few candidates correctly included all of these but the majority understood the principles covered in the question.
- Q2                      Attempted      2991              Mean      63%
- Generally this question was handled well with electrical principles of power loss and transformation well understood. The major errors in this question were caused by not including both cables in loss calculations and not stating the purpose of the third conductor although most realised it was an earth wire. Part (d) calculation of a turns ratio was generally well done but few chose to draw a well-labelled diagram or mention laminations in the iron core which is essential for efficient operation. Part (e) was very well answered with only a few inappropriately using  $V=IR$  in their explanation (large  $V$  gives large  $I$ ).
- Q3                      Attempted      2958              Mean      58%
- This surprisingly turned out to be the most difficult Section B question from the candidate's perspective. A number of candidates did the correct calculation for part (a) but then left part (b) blank, not realising it was the same calculation but with different values. The correct transition states were generally not chosen in parts (a) and (b) which is a cause for concern as this is fundamental to understanding fluorescence. In part (c) candidates did not give good Physics reasons for why UV radiation was damaging and many did not interpret the graph correctly but this was difficult for candidates who had incorrect values from the previous questions.
- Q4                      Attempted      2973              Mean      80%
- This proved to be the easiest problem solving question. The operation of open pipes was well understood and calculations were successfully carried out.
- Q5                      Attempted      2975              Mean      75%
- All parts of this question were well answered by the majority of candidates with the majority of errors related to not using the correct vertical or horizontal components. Despite an error on the paper in part (d), with the values for trajectories A and C transposed, the candidates answered the question well. As part (d) could be answered using either horizontal or vertical components candidates did not realise there was an error. Although the differing methods gave different answers, both were accepted.

- Q6                      Attempted      2939              Mean      64%  
 Parts (a) and (b) were well answered with the main errors being the conversion of units and algebraic transposition. Part (c) was poorly done with very few candidates setting out a clear and logical derivation. Many candidates realised that  $v = 2\pi r/T$  was needed but could not successfully put this in the given equation to eliminate  $r$ .
- Q7                      Attempted      2962              Mean      66%  
 Parts (a) and (b) were generally well done but many candidates did not give a clearly different second difference between the hearing of goldfish and dolphins. Part (c) was often not attempted or if attempted working was poorly set out with some attempt made to calculate intensities but the inverse square law was generally applied incorrectly.

### SECTION C: *Comprehension and Interpretation*

- Q1                      Attempted      2976              Mean      58%  
 Much more guidance was given to candidates in this year's graphical analysis question which saw an overall improvement in the handling of this type of question. Candidates continue to be poor at writing in units for constants and giving appropriate number of significant figures for data manipulation and final values. Less than 5% of candidates realise that a gradient has units, which are given by the ratio of the units on the two axes. Less than half the candidates used the gradient correctly to obtain a value for Young's modulus of the wood. Many tried unsuccessfully to plug numbers into the equation. Candidates generally showed a good understanding of what was meant by elastic when referring to materials.
- Q2                      Attempted      2970              Mean      60%  
 Parts (a) and (b) involved the same calculation,  $g = GM/r^2$  but with appropriate values. This was done well in (a) but fairly poorly in (b). Candidates used all sorts of values for the radius. Although part (c) only involved subtracting the previous two values many candidates did not do this and most forgot to include a direction which in this case is very important. Candidates did not answer part (d) well with generally non specific reasons given and did not address the question of why the surface rose and fell dramatically. Part (e) required candidates to calculate an average value for the surface gravity which was not done that well but other values were substituted appropriately. The last part required candidates to use the data to choose the correct orbital configuration. This was choice 2 but candidates obtained part marks for choice 4. Generally candidates used the information to come up with the correct or part correct answer.

### ***Issues for consideration***

No major difficulties were faced. However some areas of concern may be highlighted.

- The issue of **significant figures** seems to be worsening. Candidates do not appreciate the importance of significant figures and the information they convey about the accuracy of the data or the final value calculated. Many gave a completely inappropriate number of significant figures to estimate questions and there is still this desire to quote answers to two decimal places. This shows little understanding as the number of decimal places changes depending on the prefix chosen for the unit.
- There was some discussion about Q5 Section A with regards to it being outside the syllabus as the question used a non perpendicular magnetic field from the Earth. This seems to be an artificial restriction as the candidates are expected to calculate perpendicular components in all other questions and if this question was split in parts where the perpendicular field for the Earth was asked first then the EMF calculated from this result the question would fit the syllabus. The final result is the same and so it was decided to keep the question. It consequently turned out to be the easiest question on the paper which indicates that the candidates are comfortable with using this Physics application.

Bill Biffin  
December 2007

### **2007 Examining panel**

Chief examiner: Shelley Yeo  
Deputy: Ralph James  
Third member: Sarah Harrison

Chief marker: Bill Biffin





# **Tertiary Entrance Examination, 2007**

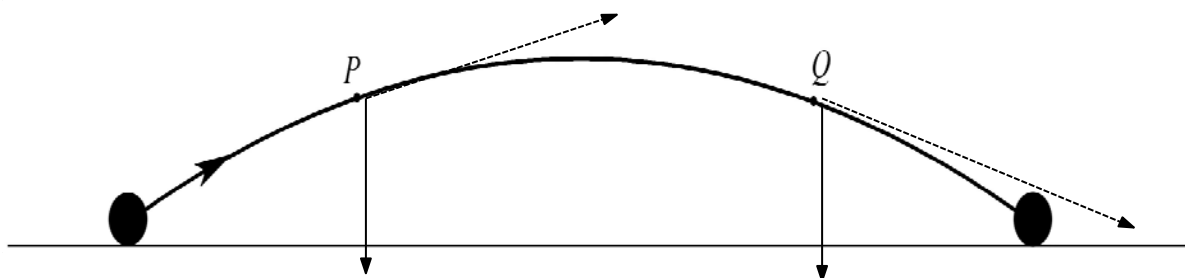
## **PHYSICS**

### **Marking Guidelines**

## SECTION A: Short Answers

(60 Marks)

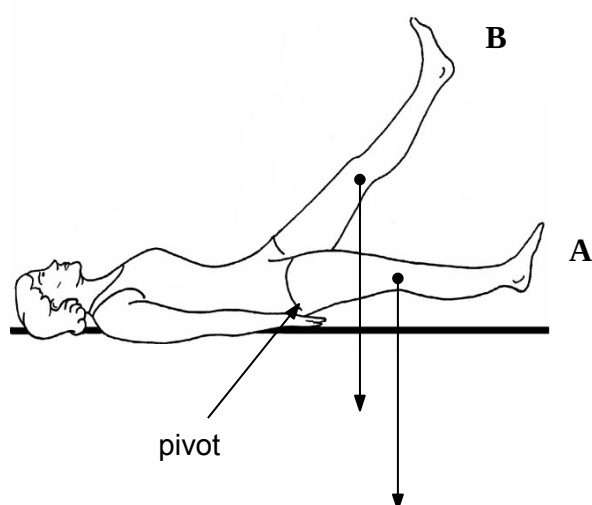
1.



- (a) 1 mark per dotted arrow  
(b) 1 mark per solid arrow

[4 marks]

2.



At (A) the torque (or CWM) provided by weight of leg is greater than at (B) due to greater horizontal distance from pivot (hip).

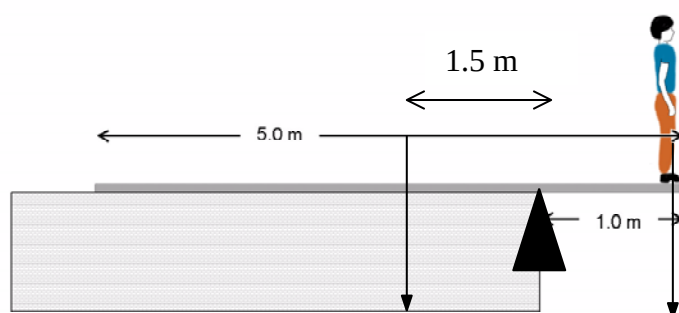
[3 marks]

Hence tension in muscle to provide balancing torque will be greater in (A) than in (B).

[1 mark]

Must refer to/use diagram for full marks

3.



Estimate mass of boy =  $70 \pm 40$  kg  
mark]

[1

Taking moments about the end of the wall  
mark]

[1

$W_{\text{plank}} \times 1.5 = W_{\text{boy}} \times 1$  therefore  $W_{\text{plank}} = 50 \times 9.8 \div 1.5 = 327$  N

Mass of plank =  $327 \div 9.8 = 33$  kg (Mass of plank =  $2/3$  Mass of boy )  
marks]

[2

4.

Radius of orbit =  $(2.0 \times 10^7 + 6.37 \times 10^6)$  m =  $2.64 \times 10^7$  m

[1

mark]

$T = (12 \times 3600)$  s

=  $4.32 \times 10^4$  s

$$v = 2 \pi R \div T$$

$$= 2 \pi (2.0 \times 10^7 + 6.37 \times 10^6) \div (12 \times 3600) = 3.84 \times 10^3 \text{ ms}^{-1}$$

[3]

**marks]**

Alternate:  $v = \sqrt{(GME \div R)}$

$$= 6.67 \times 10^{-11} \times 5.98 \times 10^{24} \div (2.0 \times 10^7 + 6.37 \times 10^6) = 3.89 \times 10^3 \text{ ms}^{-1}$$

5.

Vertical component of magnetic field =  $(2.70 \times 10^{-5} \text{ T}) \sin 65^\circ$

$$= 2.45 \times 10^{-5} \text{ T}.$$

[1 mark]

$$emf = lvB$$

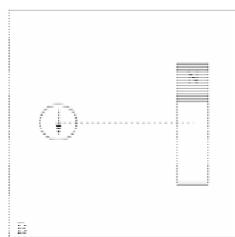
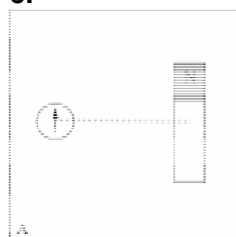
[1 mark]

$$= (59.5 \text{ m}) \times 270 \text{ m s}^{-1} \times (2.45 \times 10^{-5} \text{ T})$$

[2 marks]

$$= 0.393 \text{ V}.$$

6.



B is correct

[1 mark]

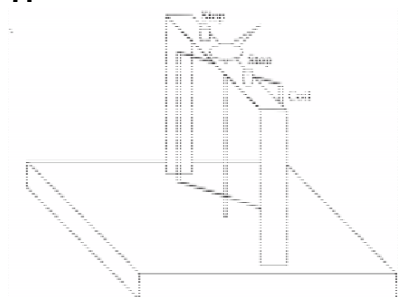
The compass needle is a magnet  
(arrow head is N-pole).

[1 mark]

It will align with the field of the bar  
magnet—and point in the direction of  
the field lines (N to S)

[2 marks]

7.

For  $180^\circ$ 

$$\Phi = BA$$

[1 mark]

$$= (1.2 \times 10^{-5}) (0.30)^2$$

$$\Delta \Phi = 2 \times (1.2 \times 10^{-5}) (0.30)^2$$

[3 marks]

$$= 2.16 \times 10^{-6} \text{ Wb}$$

8.

Wave with large time period:

$$\text{Period} = 1559 \text{ ms} - 1549 \text{ ms} = 10 \text{ ms}$$

[2 marks]

$$f = T^{-1} \therefore \text{frequency} = 100 \text{ Hz}.$$

Wave with smaller time period completes 10 whole oscillations in 10 ms, so

[2 marks]

frequency is 1000 Hz

9.

**Wave nature**

Adequate description of

1. double slit experiment OR
2. diffraction (wave fronts), OR
3. refraction (wave fronts), OR
4. polarization

[2]

**marks]****Particle nature**

Adequate description of

1. reflection (particles) OR
2. photoelectric effect ( $E = h\nu$ )
3. Compton scattering (momentum transfer)

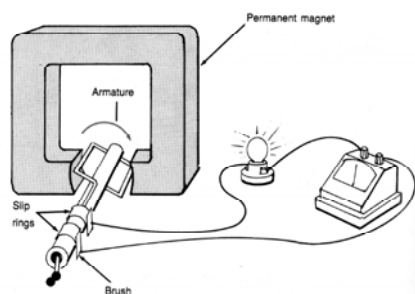
[2]

**marks]**

10.

# MARKING GUIDELINES

# PHYSICS



(a) Alternating current generator or AC dynamo

[2]

marks]

(b) Reasons

- current is produced (lighted globe),
- slip rings for alternating current
- no power source
- coil spinning between magnetic poles

[2]

marks]

11.

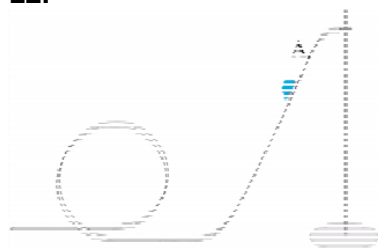
(a) The generated emf reverses because the change in flux producing the emf increases as it approaches the coil and decreases as the magnet passes through the coil. By Lenz's Law a current is produced to oppose the change in flux.

[2 marks]

(b) The emf generated is proportional to the rate of change of magnetic flux and this is proportional to the speed of the magnet. The magnet is accelerating due to gravity so the second peak is bigger.

[2 marks]

12.



Radius of loop =  $r$ ,

At top of loop:

$$mg = mv^2 \div r \quad \therefore v_{\text{top}} = \sqrt{rg} = 1.98 \text{ ms}^{-1}$$

From conservation of energy

$$PE_A = KE_{\text{top loop}} + PE_{\text{top loop}}$$

$$mgA = \frac{1}{2} mrg + mg2r$$

$$\therefore A = \frac{1}{2} r + 2r$$

$$\text{If } r = 0.4 \text{ m, } A = 1.0 \text{ m.}$$

[1 marks]

[2 marks]

[1 mark]

13.

$$130 \text{ W} = 130 \text{ J s}^{-1}$$

85% efficiency means power output is  $0.85 \times 130 = 110.5 \text{ J s}^{-1}$

[1 mark]

$$E_{\text{photon}} = hc \div \lambda$$

$$= (6.63 \times 10^{-34}) (3 \times 10^8) \div (5.90 \times 10^{-9})$$

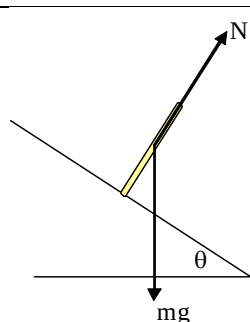
$$= 3.37 \times 10^{-19} \text{ J.}$$

[2 marks]

$$\text{Photons per second} = 110.5 \div 3.37 \times 10^{-19} = 3.28 \times 10^{20}$$

[1 mark]

14.



Derive equation from diagram:

$$\text{Vertically: } N \cos \theta = mg$$

$$\text{Horizontally: } N \sin \theta = mv^2 \div r$$

$$\therefore \tan \theta = v^2 \div rg$$

$$= (15.5)^2 \div (35 \times 9.8)$$

$$= 0.7$$

$$\text{so angle of bank} = 35^\circ$$

[2 marks]

[1 mark]

[1 mark]

15.

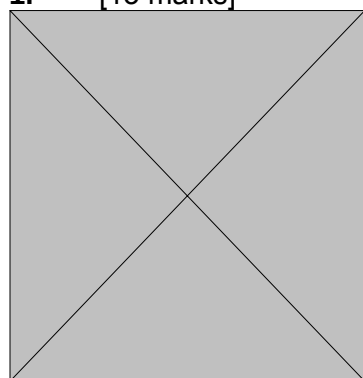
When the motor is turning at full speed, an emf is induced in the opposite direction to the applied current (Lenz's Law). The motor therefore draws a relatively small current.

[2 marks]

The jammed piece of wood stops the motor turning, there is little or no 'back' emf hence a larger current than normal flows.

[2 marks]

**1.** [13 marks]



**(a)** [3 marks]

Forces acting downwards = weight of swing + weight of baby

[1]

$$= 3 \text{ kg} + 3.6 \text{ kg} = 6.6 \text{ kg}$$

$$= 64.7 \text{ N}$$

[1]

Therefore the tension in one arm is 32.3 N

[1]

[3 marks]

**(b)** [3 marks]

$$v = s \div t$$

$$= 0.90 \div 1.6 = 0.563 \text{ ms}^{-1}$$

[1]

$$a = v^2 \div r = (0.563)^2 \div 0.60$$

$$= 0.527 \text{ ms}^{-2}$$

[2]

**(c)** [3marks]

$$\text{Total force} = mg + mv^2 \div r$$

[1]

$$= 6.6 (9.8) + 6.6 (0.563)^2 \div 0.6$$

$$= 68.2 \text{ N}$$

[1]

$$\text{Tension in each arm} = 34.1 \text{ N}$$

[1]

**(d)** [4 marks]

$$\text{stress} = \frac{F}{A} = \frac{mg + mv^2/r}{\pi r^2}$$

[1]

$$\text{stress} = \frac{(15+3)/2 * (9.8+0.53)}{\pi (7.5 \times 10^{-3})^2} = 5.26 \times 10^5 \text{ N m}^{-2}$$

[2]

This is less than the breaking stress of PVC so the swing is unlikely to break.

[1]

**2.** [16 marks]

**(a)** [2 marks]

Ground or earth line for safety reasons

[2]

**(b)** [3 marks]

$$P = VI$$

$$I = \frac{P}{V} = \frac{346 \times 10^3}{240}$$

[2]

$$= 1.44 \times 10^3 \text{ A}$$

[1]

**(c)** [4 marks]

$$R = \frac{543}{10} \times 1.74 \times 10^{-5} = 9.45 \times 10^{-4}$$

$$P = I^2 R = (1.44 \times 10^3)^2 9.45 \times 10^{-4}$$

[2]

$$= 1.96 \times 10^3 \text{ W}$$

$$\text{Total power loss} = 2 \times 1.96 \times 10^3 \text{ W} = 3.93 \times 10^3 \text{ W}$$

[1]

**(d)** [4 marks]

Diagram or description should indicate laminated iron transformer core with one turn on the secondary (240 V output) for every 275 turn on the primary (66 kV input)

[4]

Working:  $\frac{V_s}{V_p} = \frac{N_s}{N_p}$  so  $\frac{N_s}{N_p} = \frac{240}{66 \times 10^3} = 0.00363 = \frac{1}{275}$

**(e)** [3 marks]

Power is transmitted at higher voltage because this means lower current ( $P = VI$ ).

[2]

Power loss as heat generated in the wires is given by  $I^2R$  so it is minimised if the current is lower.

[1]

**3.** [15 marks]**(a)** [4 marks]

Largest wavelength:  $\Delta E = -0.55 - (-3.85) = 3.30 \text{ eV}$

[2]

$$= 3.30 \times 1.6 \times 10^{-19} = 5.28 \times 10^{-19} \text{ J}$$

$$E = hc \div \lambda$$

$$\lambda = hc \div E$$

1]

$$= (6.63 \times 10^{-34})(3 \times 10^8) \div 5.28 \times 10^{-19} = 3.77 \times 10^{-7} \text{ m}$$

1]

**(b)** [5 marks]

Req/d line corresponds to transition from level 2 to 1

1]

$$\Delta E = -2.05 - (-3.85) = 1.80 \text{ eV}$$

2]

$$= 1.80 \times 1.6 \times 10^{-19} \text{ J} = 2.88 \times 10^{-19} \text{ J}$$

$$E = hc \div \lambda$$

$$\lambda = hc \div E = (6.63 \times 10^{-34})(3 \times 10^8) \div 2.88 \times 10^{-19} = 6.91 \times 10^{-7} \text{ m}$$

1]

This corresponds to red—so stony coral is **red** under UV light

1]

**(c)** [6 marks]

**(i)** UV radiation has high energy and low wavelength so can cause ionisation of atoms/molecules and penetrate tissues.

2]

If this occurs in DNA (cell nuclei) in sufficient amounts, cell replication is affected leading to skin cancer (or permanently damaged cells)

1]

May give other specific examples

**(ii)** UV light with wavelengths in the range 375-400nm have greater than 50% transmission

[1]

**MARKING GUIDELINES**
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The % transmission light of wavelength 377 nm falls in the above range (i.e. is ~60% transmission) so this cover would be suitable.

2]

4. [14 marks]

(a) [3 marks]

$$\lambda = 2 \times 1.45 = 2.90 \text{ m}$$

1]

$$f = \frac{c}{\lambda} = \frac{346}{2.90} = 119 \text{ Hz}$$

2]

(b) [5 marks]

At 25° C and 0.0290 kg  $c = 346 \text{ m s}^{-1}$

At 37° C and 0.0296 kg

$$\text{speed} = 3.41 \times \sqrt{\frac{(273 + 37)}{0.0296}} = 349 \text{ ms}^{-1}$$

2]

$$f = \frac{c}{2L}$$

1]

$$= \frac{349}{2 \times 1.45} = 120 \text{ Hz}$$

2]

(c) [6 marks]

$$(i) \quad L = \frac{c}{2f} = \frac{349}{2 \times 88} = 1.98 \text{ m}$$

2]

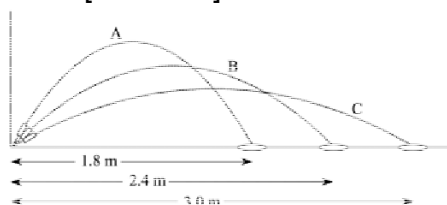
$$\text{Extra length} = 1.98 - 1.45 = 0.53 \text{ m}$$

1]

(ii) One half a metre is probably close to the distance from mouth to the lungs of a large person, so the candidate's proposition might be reasonable.

[3]

5. [16 marks]



(a) [4 marks]

$$v_H = s \div t = 2.4 \div 0.70 = 3.43 \text{ m s}^{-1} \quad [2]$$

$$v_H = v \cos 45 \quad [1]$$

$$\therefore v = v_H \div \cos 45 = 4.85 \text{ m s}^{-1} \quad [1]$$

(b) [4 marks]

$$\text{Using } t = 0.35 \text{ s} \quad [1]$$

$$S = ut + \frac{1}{2} a t^2 \quad [2]$$

$$= 4.85 \sin 45 (0.35) + \frac{1}{2} (9.8) (0.35)^2 \quad [2]$$

$$= 0.60 \text{ m (60 cm)} \quad [1]$$

# MARKING GUIDELINES

# PHYSICS

[ALT = 78 cm]

(c) [4 marks]

Wind will only affect horizontal velocity component

$$v_H = 3.43 + 0.28 = 3.71 \text{ m s}^{-1} \quad [1]$$

As vertical cpt of velocity is unchanged, t will still be 0.70 s

$$\text{Therefore } s = v t = 3.73 (0.7) = 2.6 \text{ m} \quad [2]$$

Since far edge of hole is 2.5 m away, the jet will not land in the hole. [1]

(d) [4 marks]

Working in vertical direction only,  $t = (v - u) / a$

Hence  $t \propto \Delta v$

$$\text{For A, } \Delta v (\text{vert}) = 2 \times 5.83 \sin 60 = 10 \text{ m s}^{-1}$$

$$\text{For B, } \Delta v (\text{vert}) = 2 \times 4.85 \sin 45 = 6.9 \text{ m s}^{-1}$$

$$\text{For C, } \Delta v (\text{vert}) = 2 \times 4.52 \sin 30 = 4.5 \text{ m s}^{-1}$$

Therefore if they leave at the same instant, C will land first, then B then A. [2]

Reasoning: [2]

6. [12 marks]

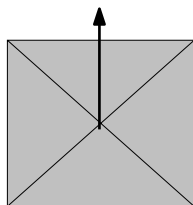
(a) [4 marks]

$$F = qvB \quad [1]$$

$$= (1.6 \times 10^{-19}) (1.10 \times 10^3) (0.200) \quad [1]$$

$$= 3.52 \times 10^{-20} \text{ N.} \quad [1]$$

Direction of force



[1]

(b) [4 marks]

$$B = \frac{mv}{qr} \quad [1]$$

$$B = \frac{1.67 \times 10^{-27} \times 10 \times 10^3}{1.6 \times 10^{-19} \times 0.015} \quad [2]$$

$$= 6.96 \times 10^{-3} \text{ T} \quad [1]$$



**MARKING GUIDELINES**
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**(c)** [4 marks]

Since  $r = \frac{mv}{qB}$  and velocity of a circular orbit is given by  $v = \frac{2\pi r}{T}$  [1]

Then  $v = \frac{rqB}{m} = \frac{2\pi r}{T}$  [1]

$T = \frac{2\pi m}{qB}$   
 $\therefore$  [1]  
 which is independent of  $r$  [1]

**7.** [14 marks]

**(a)** [3 marks]

Any 2 from:

Goldfish have maximum hearing sensitivity at a much lower frequency than dolphins. [2]

GF have a higher hearing threshold at their most sensitive frequency [1]

GF cannot hear frequencies above about 2 kHz whereas dolphins can hear to well over 100 kHz

**(b)** [5 marks]

(i)

$$\lambda = c \div f$$

[1]

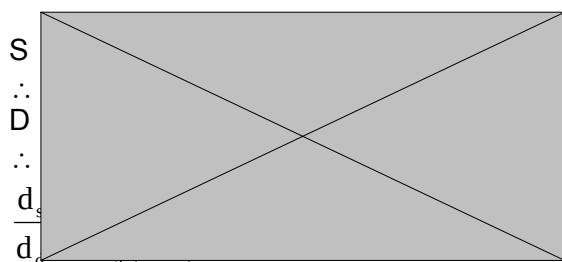
$$= 1530 \div 50 \times 10^3 = 0.0306\text{m}$$

[2]

(ii) A shad: approx 135-140 dB [1]

A dolphin: approx 40 dB [1]

**(c)** [6 marks]



had:  $140 = 10 \log (I_s \div 10^{-12})$   
 $I_s$  (minimum) =  $100 \text{ W m}^{-2}$  [1]

olphin:  $40 = 10 \log (I_d \div 10^{-12})$   
 $I_d$  (minimum) =  $1 \times 10^{-8} \text{ W m}^{-2}$  [1]

[2]  

$$d_d = \sqrt{\frac{d_s^2 I_s(\text{min})}{I_d(\text{min})}}$$

$$= \sqrt{\frac{10^2 \times 100}{1 \times 10^{-8}}}$$
  
 $\therefore$   

$$= 1 \times 10^6 \text{ m} \sim 100 \text{ km.}$$
 [2]  
 (for 135 dB,  $d = 560 \text{ km}$ )

## SECTION C: Comprehension and Interpretation

(40 Marks)

## Question 1 [20 marks]

(a) [3 marks]

$$\begin{aligned} \text{constant} &= \left( \frac{\pi^2 W T^3}{12} \right) \\ &= \left( \frac{\pi^2 (0.0286) (0.004)^3}{12} \right) \\ &= 1.51 \times 10^{-9} \text{ m}^4 \end{aligned}$$

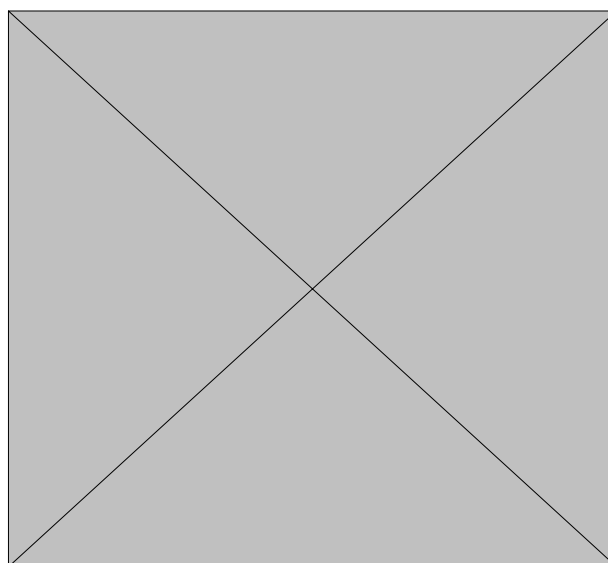
[2]  
[1]

(b) [5 marks]

| $1/L^2$ |
|---------|
| 1.00    |
| 1.18    |
| 1.49    |
| 1.98    |
| 2.60    |
| 3.84    |
| 6.25    |

Table [1]

Graph: axes [1]  
 labels [1]  
 line [2]



(c) [3 marks]

$$\begin{aligned} \text{Gradient} &= 95\text{N} \div 6.25\text{m}^2 \\ &= 15.2 \text{ N m}^2 \end{aligned}$$

[2]  
[1]

(d) [4 marks]

$$\begin{aligned} F_b &= \frac{(1.51 \times 10^{-9}) Y}{1} \times \frac{1}{L^2} \\ Y &= \frac{F_b \times L^2}{(1.51 \times 10^{-9})} \\ &= 1.0 \times 10^{10} \text{ Pa} \end{aligned}$$

[2]  
[2]

(e) [2 marks]

The wood is possibly Maple – this has the Young's modulus closest to the value determined from the graph.  
 It may also not be one of the woods mentioned as the error is approx 10%

(f) [3 marks]

It isn't correct because an elastic material is one that undergoes no permanent deformation when an applied force is removed [3]

## Question 2 [20 marks]

(a) [4 marks]

$$F = \frac{GmM_{Io}}{R_{Io}^2} = mg_{Io} \quad [1]$$

$$g_{Io} = \frac{GM_{Io}}{R_{Io}^2} \quad [1]$$

$$g_{Io} = \frac{6.67 \times 10^{-11} \times 8.94 \times 10^{22}}{1.82 \times 10^6} \quad [1]$$

$$= 1.80 \text{ ms}^{-2} \quad (\text{directed towards the centre of Io}) \quad [1]$$

(b) [3 marks]

$$F = \frac{GmM_J}{(r - R_{Io})^2} = mg_{J,inner}$$

$$g_{J,inner} = \frac{GM_J}{(r - R_{Io})^2} \quad [2]$$

$$g_{J,inner} = \frac{6.67 \times 10^{-11} \times 1.90 \times 10^{27}}{(4.22 \times 10^8 - 1.82 \times 10^6)^2} \quad [1]$$

$$= 0.718 \text{ ms}^{-2} \quad (\text{directed towards Jupiter}) \quad [1]$$

(c) [2 marks]

$$g_{inner} = 1.80 - 0.718 = 1.08 \text{ ms}^{-2} \quad (\text{towards centre of Io}) \quad [2]$$

(d) [3 marks]

The cause is related to the relative position of the moons Europa and Ganymede to Io, and the differing effects that their gravitational fields (magnitude and direction) have on Io. [1]

This has the effect of increasing the  $\Delta g$  by varying amounts over time (shown in graph). [2]

Note that the stress on Io due to Jupiter is constant and does not cause its 'tides'.

(e) [4 marks]

$$R_{Io} = 1.82 \times 10^6 \text{ m},$$

$$g \text{ (from graph)} = (12 \times 10^{-5} - 1.5 \times 10^{-5}) = 10.5 \times 10^{-5} \text{ ms}^{-2} \text{ (could accept } 12 \times 10^{-5})$$

$$g_o = (1.08 + 2.51) \div 2 = 1.8 \text{ ms}^{-2} \text{ from part (c) and extra information}$$

$$h \approx \frac{\Delta g}{g_o} R_{Io} \quad [1]$$

$$h \approx \frac{10.5 \times 10^{-5}}{1.8} 1.82 \times 10^6 \quad [2]$$

$$= 110 \text{ m} \quad (\text{if use } 12 \times 10^{-5} \text{ height is } 120 \text{ m}) \quad [1]$$

(f) [4 marks]

Answer: diagram 2 [4]